
Probability and Statistic (for CS Students)

Lecture 5: Conditional Probability and Bayes I

University of Mazandaran

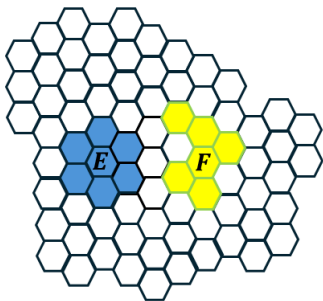
Fall 1405

Recap: Axioms of Probability

Recall: S = all possible outcomes. E = the event

- ▶ Axiom 1: $0 \leq P(E) \leq 1$
- ▶ Axiom 2: $P(S) = 1$
- ▶ Axiom 3: If events E and F are mutually exclusive:
 $P(E \cup F) = P(E) + P(F)$

Recap: Mutual Exclusive Events



If events are mutually exclusive, probability of OR is simple:

$$P(E \cup F) = P(E) + P(F)$$

Recap: Mutual Exclusive Events

$$P(E_1 \cup E_2 \cup \dots \cup E_n) = \sum_{i=1}^n P(E_i)$$

Recap: Mutual Exclusivity and The 3rd Axiom

$$P(E') = 1 - P(E)$$

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Warmup!

Say, the population of UMZ is 15,000 people

- ▶ You are friend with let's say 100?
- ▶ Walk into a room, see 260 random people.
- ▶ What is the probability that you see someone you know?
- ▶ Assume you are equally likely to see each person at UM

Say, the population of UMZ is 15,000 people

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Sometimes, it is easier to calculate $P(E')$

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- ▶ Define the sample space
- ▶ Count the size of the sample space
- ▶ Define the event
- ▶ Count the size of the event
- ▶ Calculate the probability

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rooms with 260 people in it
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 $\binom{15000}{260}$
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 $n(E') = \binom{15000-100}{260}$
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$$P(E') = \frac{\binom{15000-100}{260}}{\binom{15000}{260}} \approx 0.17$$

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$$P(E') = \frac{\binom{15000-100}{260}}{\binom{15000}{260}} \approx 0.17$$

$$P(E) = 1 - P(E') \approx 0.83$$

Conditional Probability

Another Dice Example

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Let E be the event that the sum is 4.

$$\left\{ \begin{array}{cccccc} [1,1] & [1,2] & [1,3] & [1,4] & [1,5] & [1,6] \\ [2,1] & [2,2] & [2,3] & [2,4] & [2,5] & [2,6] \\ [3,1] & [3,2] & [3,3] & [3,4] & [3,5] & [3,6] \\ [4,1] & [4,2] & [4,3] & [4,4] & [4,5] & [4,6] \\ [5,1] & [5,2] & [5,3] & [5,4] & [5,5] & [5,6] \\ [6,1] & [6,2] & [6,3] & [6,4] & [6,5] & [6,6] \end{array} \right\}$$

$E =$ In red

Another Dice Example

Roll two 6-sided dice, yielding values D_1 and D_2 .

Let E be event: $D_1 + D_2 = 4$

What is $P(E)$?

$$|S| = 36$$

$$E = \{[1, 3], [2, 2], [3, 1]\}$$

$$P(E) = \frac{3}{36} = \frac{1}{12}$$

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What is $P(E, \text{given } F \text{ already observed})$

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$$S = \{[2, 1], [2, 2], [2, 3], [2, 4], \\ [2, 5], [2, 6]\}$$

$$E = \{[2, 2]\}$$

$$P(E) = \frac{1}{6}$$

Conditional Probability

The **conditional probability** of E given F is the probability that E occurs given that F has already occurred. This is known as conditioning on F .

Written as: $P(E|F)$

Means: “ $P(E, \text{given } F \text{ already observed})$ ”

Sample space $\rightarrow$ all possible outcomes consistent with F (i.e., $S \cap F$)

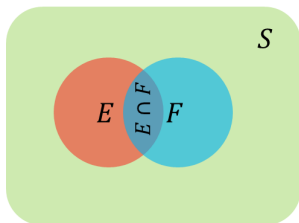
Event space $\rightarrow$ all outcomes in E consistent with F (i.e., $E \cap F$)

Visual Intuition

The **conditional probability** of E given F is the probability that E occurs given that F has already occurred. This is known as conditioning on F .

With **equally likely outcomes**:

$$P(E|F) = \frac{\# \text{ outcomes in } E \text{ consistent with } F}{\# \text{ outcomes in } S \text{ consistent with } F} = \frac{|E, F|}{|S, F|} = \frac{|E, F|}{|F|}$$



Conditional Probability In General

General definition of conditional probability:

$$P(E|F) = \frac{P(E, F)}{P(F)}$$

Rewrite (chain rule):

$$P(E, F) = P(E|F) P(F)$$

Note: these properties hold even if outcomes are not equally likely.

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What if $P(F) = 0$?

- ▶ $P(E|F)$ undefined
- ▶ Lucky you! You observed impossible!

فیلیمو



What is the probability that a user will watch “Age of Samurai: Battle for Japan”?



Filimo and Conditional Probability

What is the probability that a user will watch “Age of Samurai: Battle for Japan”?



$$S = \{\text{Watch, Not Watch}\}$$

$$E = \{\text{Watch}\}$$

$$P(E) = \frac{1}{2}$$

Filimo and Conditional Probability

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$$P(E) = \lim_{n \rightarrow \infty} \frac{\text{count}(E)}{n} \approx \frac{\# \text{ people who watched the series}}{\# \text{ people on Filimo}}$$

Filimo and Conditional Probability

What is the probability that a user will watch “Age of Samurai: Battle for Japan”?



$$P(E) = \lim_{n \rightarrow \infty} \frac{n(E)}{n} \approx \frac{\# \text{ people who watched the series}}{\# \text{ people on Filimo}}$$

$$P(E) = \frac{100000}{9.5 \times 10^6}$$

Filimo and Conditional Probability

Let E be the event that a user watches the given movies.



$$P(E) = 0.55$$



$$P(E) = 0.05$$



$$P(E) = 0.75$$



$$P(E) = 0.1$$

Let E = a user watches Shogun

Let F = a user watches Age of Samurai

What is the probability that a user watches Shogun, given they watched "Age of Samurai"?

$$P(E|F)$$

Filimo and Conditional Probability

Let E = a user watches Shogun

Let F = a user watches Age of Samurai

What is the probability that a user watches Shogun, given they watched Samurai?

$$P(E|F) = \frac{P(E, F)}{P(F)} = \frac{\frac{\text{\# people who have watched both}}{\text{\# people on Filimo}}}{\frac{\text{\# people who have watched Samurai}}{\text{\# people on Filimo}}}$$

Filimo and Conditional Probability

Let E = a user watches Shogun

Let F = a user watches Age of Samurai

What is the probability that a user watches Shogun, given they watched Samurai?

$$\begin{aligned} P(E|F) &= \frac{P(E, F)}{P(F)} = \frac{\frac{\# \text{ people who have watched both}}{\# \text{ people on Filimo}}}{\frac{\# \text{ people who have watched Samurai}}{\# \text{ people on Filimo}}} \\ &= \frac{\# \text{ people who have watched both}}{\# \text{ people who have watched Samurai}} \end{aligned}$$

Filimo and Conditional Probability

Let E be the event that a user watches the given movies.

Let F be the event that the same user watches “Blue Eye Samurai”



$$P(E) = 0.55$$

$$P(E|F) = 0.9$$



$$P(E) = 0.05$$

$$P(E|F) = 0.05$$



$$P(E) = 0.75$$

$$P(E|F) = 0.68$$



$$P(E) = 0.1$$

$$P(E|F) = 0.5$$

And

Or

Given

$$P(E \text{ and } F)$$

$$P(E \text{ Or } F)$$

$$P(E|F)$$

$$P(E, F)$$

$$P(EF)$$

$$P(E \cap F)$$

$$P(E \cup F)$$

$$P(E|F, G)$$

Conditional Probability Equation

$$P(E|F) = \frac{P(E, F)}{P(F)}$$

$$P(E, F) = P(E|F) \cdot P(F)$$

Chain Rule and Generalized Conditional Probability

For events $E_1, E_2, E_3, \dots, E_n$, the chain rule states:

$$\begin{aligned} P(E_1, E_2, E_3, \dots, E_n) &= P(E_n | E_1, E_2, \dots, E_{n-1}) P(E_1, E_2, \dots, E_{n-1}) \\ &= P(E_n | E_1, E_2, \dots, E_{n-1}) P(E_{n-1} | E_1, E_2, \dots, E_{n-2}) P(E_1, E_2, \dots, E_{n-2}) \\ &\dots \\ &= P(E_1) P(E_2 | E_1) P(E_3 | E_1, E_2) \dots P(E_n | E_1, E_2, \dots, E_{n-1}) \\ &= P(E_1) \prod_{i=2}^n P(E_i | E_1, \dots, E_{i-1}) \end{aligned}$$

Chain Rule: Example

We randomly draw 4 cards without replacement from a deck with 52 cards. What is the probability that we have picked 4 aces?

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We randomly draw 4 cards without replacement from a deck with 52 cards. What is the probability that we have picked 4 aces?

E_i : the i^{th} draw is ace

$$P(E_1) = \frac{4}{52}$$

$$P(E_2|E_1) = \frac{3}{51}$$

$$P(E_3|E_1, E_2) = \frac{2}{50}$$

$$P(E_4|E_1, E_2, E_3) = \frac{1}{49}$$

$$P(E_1, E_2, E_3, E_4) =$$

$$P(E_1) P(E_2|E_1) P(E_3|E_1, E_2) P(E_4|E_1, E_2, E_3) = \frac{4}{52} \times \frac{3}{51} \times \frac{2}{50} \times \frac{1}{49}$$

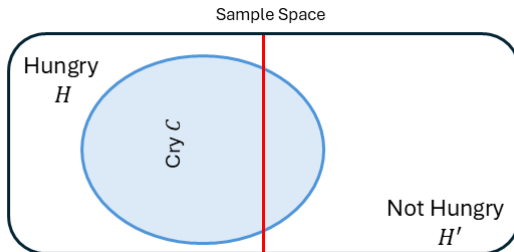
When a baby wakes up in the morning, she has a 50% chance of being hungry. The chance that the baby cries given that she is hungry is 50%.

Q1: what is the probability of crying, unconditioned?

Q2: what information do we need?

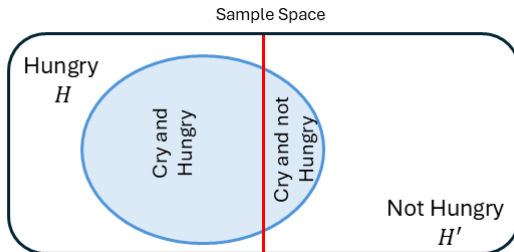
Law of Total Probability

Say C and H are events in S



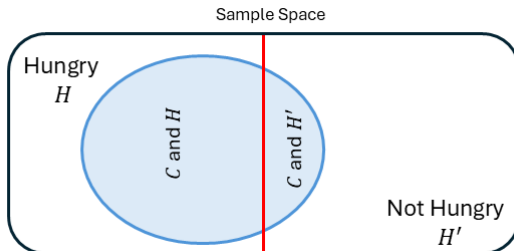
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Law of Total Probability

Say C and H are events in S



$$\begin{aligned} P(C) &= P(C, H) + P(C, H') \\ &= P(C|H)P(H) + P(C|H')P(H') \end{aligned}$$

Theorem: Let F be an event where $P(F) > 0$. For any event E ,

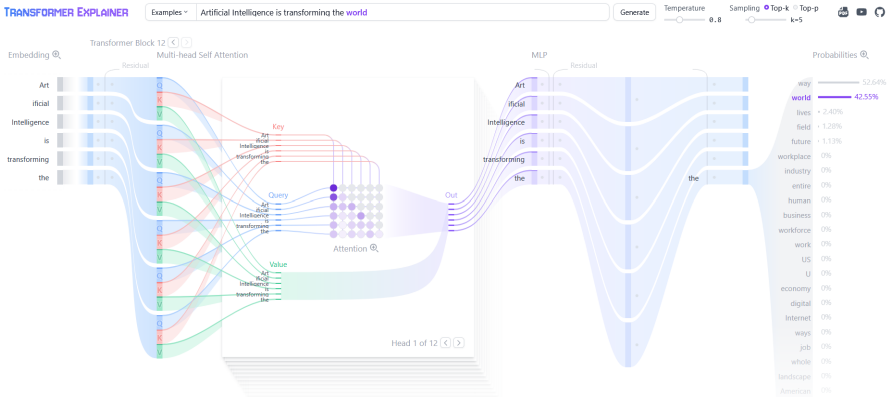
$$P(E) = P(E|F)P(F) + P(E|F')P(F')$$

Proof:

- ▶ $E = (E, F) \text{ or } (E, F')$
- ▶ $P(E) = P(E, F) + P(E, F')$
- ▶ $P(E) = P(E|F)P(F) + P(E|F')P(F')$

Conditional Probability as a Building Block of LLMs

TRANSFORMER EXPLAINER



<https://poloclub.github.io/transformer-explainer/>

Conditional Probability as a Building Block of LLMs

Let T_i be the i -th token in a prompt.

An LLM is built to compute $P(T_i | T_1, \dots, T_{i-1})$

